

Ceiling Fans:

Fulfilling the Energy Efficiency Promise



KING OF FANS

Ceiling fans, and their light kits, can be substantial energy users. Changes in product design and consumer education offer major savings opportunities.

by Chris Calwell and Noah Horowitz

Ceiling fans are one of the most popular and generally well regarded of all home energy efficiency features. They have a very pragmatic appeal to those who want to cut summer electric bills. At the same time, they offer a decorative alternative to typical light fixtures. So it should come as no surprise that two-thirds of American households now have ceiling fans. Home improvement centers do a brisk and growing business in sales to do-it-yourselfers, and it's not

uncommon to find ceiling fans installed by builders in nearly every bedroom and living area in many new homes.

But recent research suggests that it may be time for the energy efficiency community to take a second look at ceiling fans (see "Ceiling Fans 101," p. 27). The Florida Solar Energy Center identified ceiling fans as substantial energy users and investigated a number of technologies to improve energy efficiency (see "Cutting-Edge Blades Slash Fan Energy Use," *HE*,

July/Aug '99, p. 7). The Natural Resources Defense Council (NRDC) and Ecos Consulting recently undertook our own EPA-funded research to determine how much energy ceiling fans use, identify energy savings opportunities, and explore various strategies to transform this market.

Surpassing Predictions

Our research suggests that ceiling fans are using far more energy than previous research had predicted.

There are several reasons for this. Ceiling fan sales are climbing, so older studies projected lower saturation levels than we find today. Most ceiling fan research has examined motor energy use only, but we found that lighting energy use is actually greater in most regions. Finally, the part of the country with the most fans (the South) also tends to have the highest hours of usage per fan, so regional energy consumption varies dramatically from assumed national averages.

The U.S. Department of Energy's Residential Energy Consumption Survey (RECS) data draw a fairly accurate assessment of the ceiling fan population by geographic region. As expected, ceiling fan ownership tends to be highest in the warmest climates, and lowest in the Pacific Coast and New England regions (see Figure 1).

Few studies have documented average hours of operation for ceiling fan motors and lighting systems by region. Use has been measured for small samples in Florida and Canada, but far more utility research is needed to determine hours of operation with precision. To obtain a better estimate of regional and national motor energy consumption, we developed a model that correlates hours of operation with the average number of cooling degree-days (CDD) in particular census regions. Our model is calibrated with the hours of use found in the Canada and Florida studies.

We found an average of 1.5 ceiling fans per household, or 153 million fans nationwide. Average motor use is 6.2 hours per

day; lighting use averages about 3.3 hours per day. Because ceiling fan lights are often the brightest, most centrally located, and most conveniently switched fixtures in the room, average lighting use may be even higher.

This suggests that the motors of ceiling fans use about 200 kWh per household-year. While their lights use about 320 kWh

per household-year, the total—520 kWh per household-year—is greater than the annual consumption of many new energy-efficient refrigerators. In some regions, household electricity use for ceiling fans is far greater (see Figure 2). On a national basis, ceiling fans may account for as much as 5% of current residential electric use.

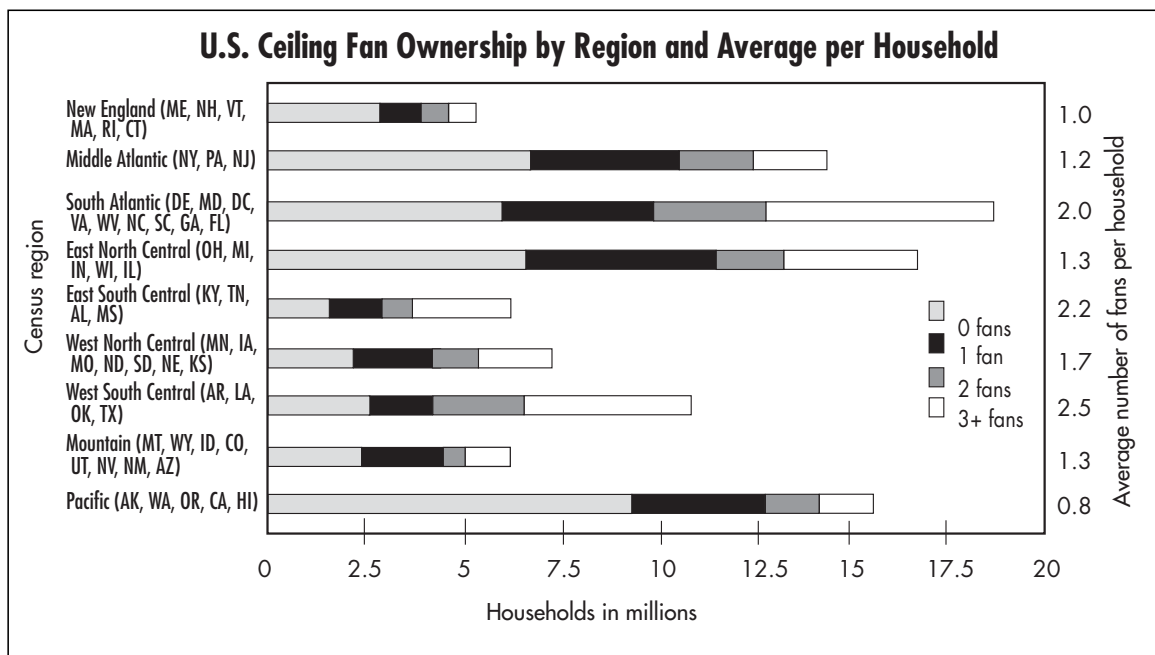


Figure 1. Fan popularity varies dramatically throughout the United States.

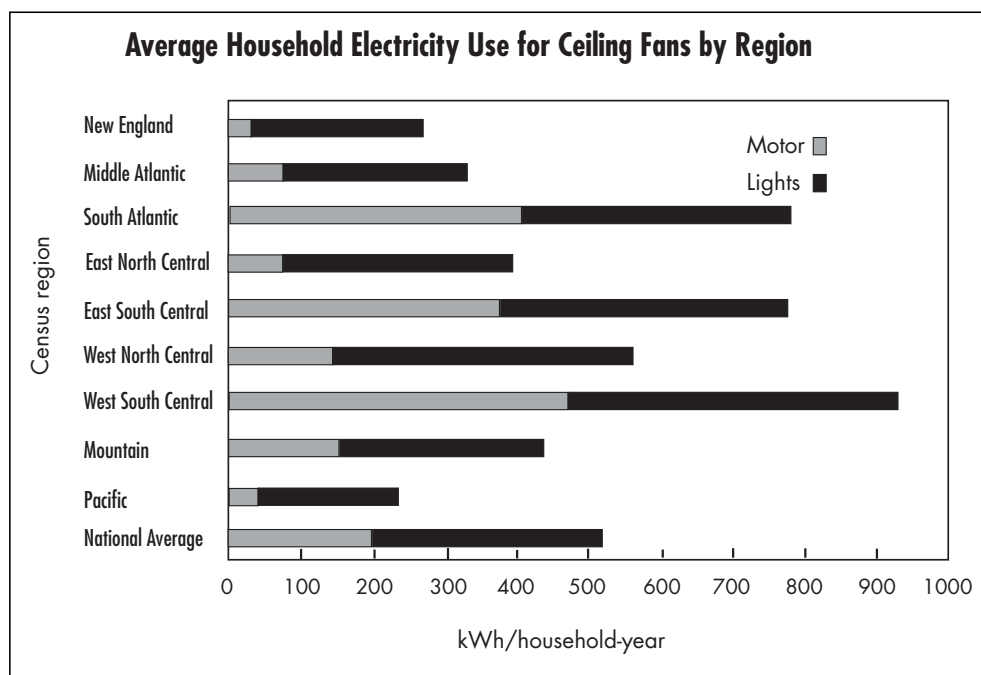


Figure 2. Light energy use is greater than motor energy use in most regions.

The Opportunity

If used efficiently, ceiling fans offer tremendous potential for energy savings. Such savings can come from energy-efficient motors, better blade aerodynamics, energy-efficient lighting, more intelligent controls, and better education of installers and purchasers. In other words, ceiling fans could provide more comfort and better light while using less energy.

The opportunity to improve motor efficiency is enormous, given that most ceiling fan motors today waste 80%–95% of their electricity as heat. However, only some of these savings can be captured in the near term with off-the-shelf products. The greatest opportunity to improve ceiling fan motor efficiency will happen over the long term, as marketing and incentive programs drive manufacturers to specify more efficient permanent-magnet motor designs from their original equipment manufacturer (OEM) suppliers. According to industrial efficiency expert Neal Elliott of the American Council for an Energy-Efficient Economy (ACEEE), ceiling fan motor efficiencies could be doubled or tripled by increasing typical motor cost from a mere \$6 to \$12 per unit.

Recent research suggests that air flow can be increased by at least 40% at a given power level (or power level reduced by 40% for a given air flow) when flat blades are replaced with curved blades similar to propellers. This design adds little cost if the blades are made of molded materials such as plastic, but costs could rise significantly if the blades are made of wood or metal. Improvements to air flow per watt can be achieved by either motor or blade improvements, and individual manufacturers are looking for the most cost-effective and appealing ways to combine the two.

Using compact fluorescent lamps (CFLs) could make ceiling fan lighting even more energy-efficient. CFLs offer four advantages in ceiling fan applications:

- Long life reduces the hassle of changing bulbs, especially in rooms with high ceilings.
- Energy savings may be even greater than the savings from better motors and blades.
- Light output can be much higher, given the UL limit of 60 watts per socket imposed on most light kit designs to prevent overheating.
- Lower heat output improves occupant comfort in most applications.

However, there are also challenges associated with using screw-based CFLs in existing ceiling fans. First costs are higher, of course, though annual bulb and electricity costs are significantly lower. The globe style sold in most ceiling fan lighting kits often makes it difficult or unattractive for the user to install CFLs. In addition, motor vibration may shorten the lifetime of some CFLs, though more data are needed to verify that problem.

The use of multiple bulbs in ceiling fans compounds energy waste. Four 60W incandescent bulbs produce less light than a single 200W bulb, just as four 15W CFLs produce less light than a single 60W CFL. In addition, the small incandescent bulbs specifically marketed for ceiling fans are even less efficient than standard incandescent bulbs of the same wattage. This makes it difficult to get enough light, especially if the fan is mounted on a high ceiling.

Using a single, high-output compact fluorescent source offers a number of intriguing advantages (see Table 1). Most importantly, it offers very high light output with maximum convenience at the lowest life cycle cost. Making it a dedicated Energy Star fixture with a hard-wired ballast ensures that the energy savings will persist. In other words, a pin-

based socket prevents the consumer from inserting a screw-based incandescent bulb into the fixture. The only bulb that can fit in the socket is an energy-efficient one.

Improvements to controls also offer substantial energy savings, by more closely matching fan operation to room conditions and occupancy. Simply placing motor speed and direction controls in a convenient location accomplishes part of this goal. Beyond that, a number of control technologies are possible. Some utilize infrared or motion-sensing technology to switch off fans after people leave the room. Others employ timers. Still others match fan speed automatically to air temperature.

Educating fan users may be even more important. There is a widespread belief that ceiling fans automatically save energy or otherwise “freshen” the air. This leads many people to switch them on at the beginning of the warm season and leave them running continuously for months (see “Getting the Most from Your Fan,” p. 28). Better consumer education about ceiling fan operation could encourage people to switch them off when they leave the room, just as they know they should do with the lights.

What Next? Policies and Strategies

An important transition has begun with ceiling fans. Once thought of primarily as pieces of rotating furniture whose only differences were aesthetic, ceiling fans are now beginning to be seen as air-moving and room-illuminating devices with widely differing performance characteristics. Manufacturers have always promoted the range of wood and metal finishes available on their fans, but today, issues like motor warranty, operating noise, and measured air flow are becoming

Table 1. Comparing the Various Options for Ceiling Fan Lighting

Bulb Type and Number	Total Power Use (Watts)	Total Light Output (Lumens)	Efficiency (Lumens/Watt)	Bulb Life (Hours)	KWh/Yr	Annual Cost* (Bulbs and Electricity)
60W incandescent (4)	240	3,460	14.4	1,000	289	\$25.54
60W ceiling fan incandescent (4)	240	2,600	10.8	1,500	289	\$26.34
150W halogen (1)	150	2,800	18.7	2,000	181	\$16.86
20W CFL (4)	80	4,080	51.0	10,000	96	\$13.49
60W 2D or 2C (1)	60	4,000	66.7	12,000	71	\$7.47

* Assumes an average electricity rate of 8¢ per kWh.

increasingly important too.

A focus on energy efficiency is a natural extension of this interest in performance. NRDC and Ecos Consulting are currently working with the U.S. Environmental Protection Agency and numerous manufacturers to draft an Energy Star specification for ceiling fans. This specification would define a consensus method for testing air

flow and noise that would make it possible to compare the performance of different products accurately. Fans would need to move a specified CFM per watt at each of their three operating speeds, operate below a specified noise threshold, and incorporate Energy Star-compliant lighting.

Manufacturer King of Fans will introduce improved blade design and ded-

icated pin-based compact fluorescent lights in a Hudson Bay product available at Home Depot in 2001 (see photo on cover). The arrival of an Energy Star labeling program shortly thereafter should help draw attention to efficient products already on the market, while prompting other manufacturers to make needed improvements to existing products.

Ceiling Fans 101

It is helpful to think of a ceiling fan as a system consisting of four key parts: the motor, the fan blades, the lights, and the controls. This system can be mounted directly to the ceiling, or suspended a few inches to a few feet below the ceiling on an extension pole. (The latter increases efficiency by increasing available air flow above the blades.) The motor is mounted vertically in the middle of the fan. It is usually designed to rotate the blades both clockwise and counterclockwise at one of three user-selectable speeds.

The motors in ceiling fans typically consume about 60 to 100 watts at high speed, 20 to 40 watts at medium speed, and 10 to 15 watts at low speed.

These motors are optimized for longevity and quiet operation, but they are not usually optimized for energy efficiency—a problem common to fractional horsepower AC motors (see “Motors Matter,” HE, July/Aug ’00, p. 31). Though small-motor efficiencies are notoriously difficult to measure with precision, it appears that ceiling fan motor efficiencies today average about 5%–20%—dramatically lower than the efficiencies of the larger motors governed by federal efficiency standards. These standards don’t apply to fractional horsepower motors like the ones found in ceiling fans.

There are usually three to five fan blades connected to the motor. As with wind turbines, the fewer the blades, the greater the aerodynamic efficiency. However, the

overwhelming majority of U.S. buyers prefer the five-bladed model for aesthetic reasons. The blades are usually flat pieces of wood, plywood, metal, or plastic, mounted at a downward-facing angle. While such blades are inexpensive to manufacture, they are far



This ceiling fan is typical of the five-blade design consumers prefer.

less effective at moving air than the curved designs found in propellers and most other fan types.

The simple, flat blades do a very poor job of converting the rotational motion to downward air flow. Instead, some 90% of the rotational energy is wasted as turbulence, lateral air flow, or wind noise. The combined efficiencies of the motor and blades lead to a startling conclusion: only about 1%–2% of the electricity that goes into the motors actually imparts downward motion to air. By contrast, a well-designed wind turbine can turn more than 25% of the wind power that passes through it into electricity. Fortunately the fan industry is now moving rapidly to improve the blade design. Products with curved, more efficient designs

are expected to hit the market in early 2001.

According to a recent study of existing homes throughout California, sponsored by the San Diego Gas and Electric Company, about 95% of all ceiling fans include lights. The lights may be either integrated into the

bottom of the fan or provided as a separate kit to be connected later. Ceiling fans frequently include three to five individual light sockets with decorative globes, each fitted with an incandescent bulb. The trend in recent years is toward simpler designs with a single light source mounted in a globe in the center of the fan’s hub, normally with a higher-wattage

halogen or incandescent source.

Like the motor and blade systems, these incandescent lamps waste energy. They convert more than 90% of their electricity input immediately into heat, confounding the product’s design objective of cooling the user.

The controls include switches for motor speed, motor direction, and lights. Most designs allow on-off control of the motor and lights through wall switches, but motor speed and direction are normally controlled through switches mounted on the fan itself. More recent designs consolidate those functions in a remote control or more full-featured wall switch, making it far more convenient for the user to control fan operation precisely.

Getting the Most from Your Fan—Tips for Maximizing Energy Savings

Do a little comparison shopping. Not all ceiling fans are created equal. The energy consumption of a ceiling fan can be greatly reduced by using efficient motors and optimized blade designs. The best models are 3 to 4 times as efficient as the worst models, and tend to be quieter as well (see Figure A). Look for models with the highest air flow (CFM) and the lowest power use (watts) at each of their three speeds.

However, each manufacturer uses a different test method to measure air flow, so the results are more useful within a company's product line than across different companies. Home Depot stores currently provide free brochures listing the air flow and power use of Hunter and Hampton Bay fans. Most other manufacturers do not yet publish air flow and power figures, making it difficult to assess performance.

Buy an energy-efficient light kit for your fan. If your ceiling fan is the primary light fixture in a heavily used room, you may save more energy by upgrading the lighting than by getting a better fan. Light kits are often sold separately from ceiling fans, and their globes are frequently too small to accommodate CFLs.

In 2001, Energy Star-labeled light kits should become available for ceiling fans. These kits come with hard-wired ballasts and pin-based CFLs. They cut lighting electricity use by up to 75% and often provide brighter lighting levels than the multiple 60W incandescent bulbs typically used. They also offer far greater convenience, running for years before they need replacement.

Fans are for people, not rooms. Room air temperatures are normally 20°F–30°F cooler than body temperature. Ceiling fans can improve occupant comfort at a given room temperature by creating a windchill effect when they blow relatively cool air across the skin. Because the furniture and the thermostat are already at roughly the same temperature as the room air, they can't "feel" the windchill. So when you leave the room, switch the fan and its light off—you'll save energy and money.

For every season, turn, turn, turn.... During the summer, a ceiling fan increases comfort by blowing air downward on room occupants. But that same windchill can actually feel like a draft in winter, which is why ceiling fan motors should be reversed to blow air upward in winter (see Figure B, p.29). This helps disperse the warm air that tends to gather near ceilings, distributing it more evenly throughout the room, especially around the perimeter and near the floor. Fan direction is normally controlled by a slide switch on the motor housing. This may be a little inconvenient to reach, but throwing that switch twice a year will pay for itself in improved comfort and energy savings.

Dial in the savings with thermostats. Ceiling fans can save quite a bit of energy if summer users raise air conditioning thermostats. In the Florida climate, these net savings are roughly 14% for a 2°F increase,

according to the Florida Solar Energy Center. The reason is no big mystery; ceiling fan motors use only 10 to 100 watts, depending on size and speed.

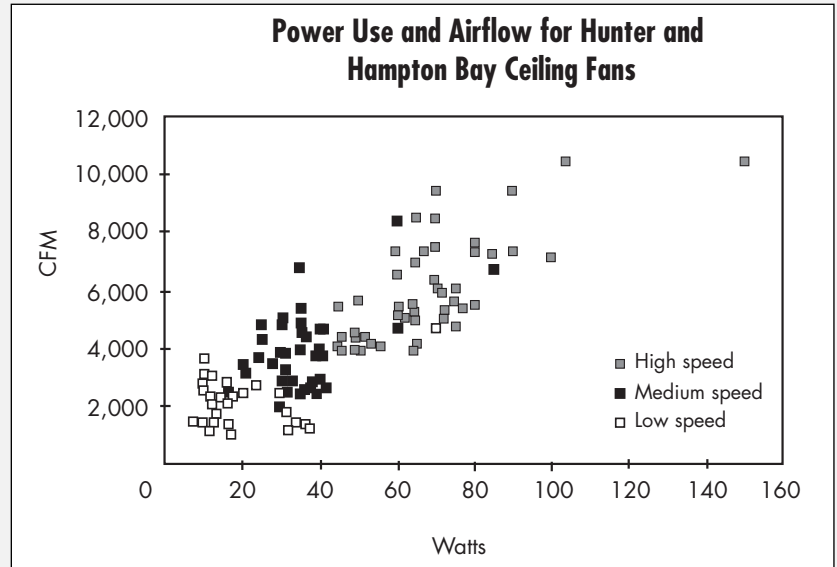


Figure A. For any given air flow, ceiling fans consume a proportional amount of energy.

By contrast, central air conditioners gobble 2,000 to 5,000 watts, while room air conditioners consume 600 to 3,000 watts.

Savings are also possible on the heating side. Electric space heaters typically devour 600 to 1,500 watts per room. So any time you can run a ceiling fan instead of one of these guzzlers to achieve the same level of comfort, you are saving energy.

But the trick is in the thermostat. The Florida Solar Energy Center found no difference between the air conditioner thermostat settings in Florida houses that had ceiling fans operating and those that didn't. As a result, the houses that were using ceiling fans had even higher energy bills!

The lesson is simple: Give the thermostat dial a spin if you want to save energy. You may find that ceiling fans are the perfect way to extend the shoulder season as well—shortening the number of days you need air conditioning in the summer and central heating in the winter.

Know your limits. Ceiling fans can supplement traditional HVAC systems, but cannot duplicate the wide range of functions they perform: ventilation, filtration, humidification, dehumidification, heating, and cooling. Unlike whole-house fans or even box fans in an open window, ceiling fans are not ventilation devices; they are circulation devices. They don't bring cooler air into a home or vent warm, humid air from it.

Though some ceiling fans now include filters within their blades to trap particles and odors, most have no real effect on the relative staleness of air in a room. The way to get rid of stale air is to exhaust it from the room and bring in fresh or filtered air to replace it. And don't count on ceiling fans to cool things off by themselves on a really hot day; if the air in the room becomes as warm as body temperature, that "breeze" won't provide much windchill.

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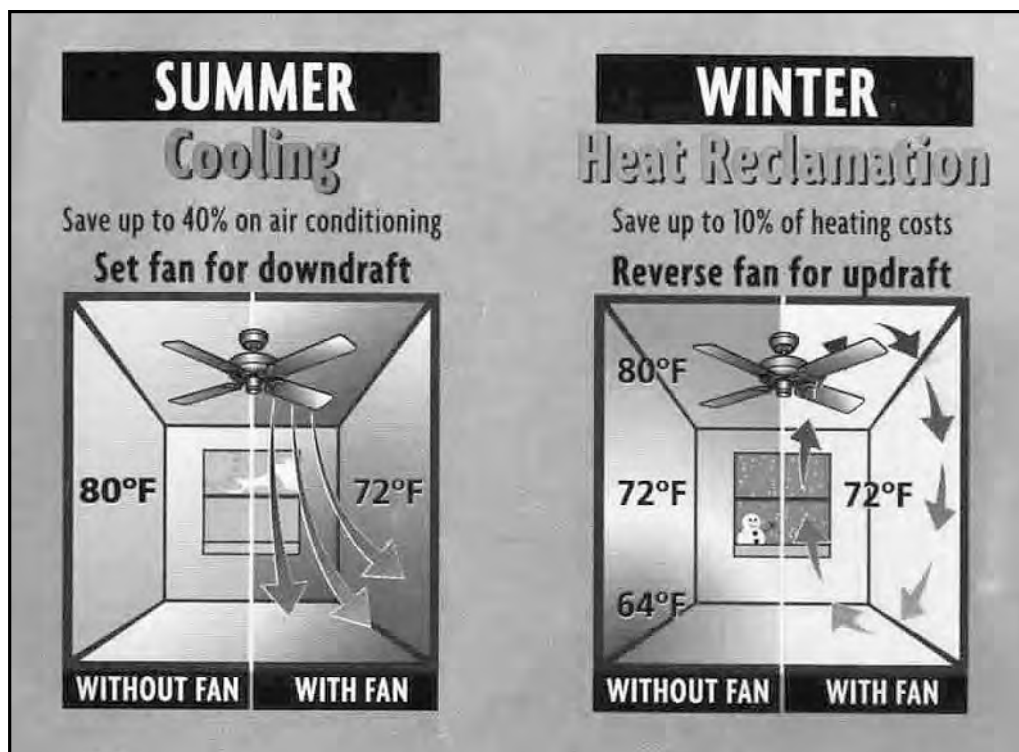


Figure B. In summer, a fan can reduce air conditioning bills by up to 40%. In winter, by reversing the direction of the fan and reclaiming hot air trapped at the ceiling, heating costs can be lowered by as much as 10%.

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At the same time, there remains a need for better data from utilities and manufacturers on typical hours of use for ceiling fan motors and lights. These studies will help improve estimates of likely energy savings that can be achieved from Energy Star technology, and also guide utility efforts to reduce summer peak demand. NRDC is working with utilities to encourage additional studies of this type in a variety of regions.

The marketing infrastructure for efficient ceiling fans will also need to grow to facilitate comparisons among competing products. It may no longer be enough to hang the fans out of reach among dozens of competing models; discerning customers will want to see for themselves how one fan's air flow, light output, and noise compare to another's.

As utility programs emerge to support Energy Star ceiling fans, the same kinds of brand-building efforts that have occurred with appliances, office equipment, and lighting will also come into play. Energy Star-labeled products will be more expensive than standard products, so utility programs may need to provide financial incentives, salesperson training, merchandising, point of purchase displays, and other tools that have been effective in encouraging the sale of other Energy Star products. Many of these programs may need to focus specifically on builders and contractors, recognizing that the market for ceiling fans is broader than simply homeowners and do-it-yourselfers.

Finally, a strong emphasis should be placed on consumer education, not just in selecting a fan, but in operating it wisely. Teaching consumers how to use ceiling fans efficiently will not be an easy job. The process can begin with more consistent and clear information on the package and in the operating manual. But it will also require spreading the word to others who influence ceiling fan operation, including builders, installers, salespeople, utilities, and consumer education specialists.

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